Homework 5

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1. If G is a finitely-generated abelian group, G is isomorphic to a direct product of cyclic groups. However, the decomposition as a product of cyclic groups is not in general unique.

(a) Find an element of order 30 in the group $\mathbb{Z}/5\mathbb{Z} \times \mathbb{Z}/6\mathbb{Z}$.

(b) Use your answer to (a) to prove that $\mathbb{Z}/5\mathbb{Z} \times \mathbb{Z}/6\mathbb{Z} \cong \mathbb{Z}/30\mathbb{Z}$.

Proof.

(a) The element $(1, 1)$ has order 30.

(b) Since the order of $\mathbb{Z}/5\mathbb{Z}\times\mathbb{Z}/6\mathbb{Z}$ is 30, part (a) shows that it is cyclic since it is generated by one element. Thus $\mathbb{Z}/5\mathbb{Z} \times \mathbb{Z}/6\mathbb{Z} \cong \mathbb{Z}/30\mathbb{Z}$ since cyclic groups of the same order are isomorphic.

$\square$

2. Let C be a non-singular cubic curve given by $y^2 = f(x) = x^3 + ax^2 + bx + c$.

(a) Prove that

$$\frac{d^2y}{dx^2} = \frac{2f''(x)f(x) - f'(x)^2}{4yf(x)} = \frac{\psi_3(x)}{4yf(x)}.$$

Use this to deduce that a point $P = (x, y) \in C$ is a point of order 3 if and only if $P \neq \mathcal{O}$ and P is a point of inflection on the curve C.

(b) Suppose now that $a, b, c \in \mathbb{R}$. Prove that $\psi_3(x)$ has exactly two real roots, say α_1 and α_2 with $\alpha_1 < \alpha_2$. Show that $f(\alpha_1) < 0$ and $f(\alpha_2) > 0$. (From this it follows that $C[3](\mathbb{R}) \cong \mathbb{Z}/3\mathbb{Z}$.)

Proof.

(a) Indeed,

$$\begin{aligned} f(x) &= x^3 + ax^2 + bx + c = y^2 \\ f'(x) &= 3x^2 + 2ax + b = 2y\frac{\mathrm{d}y}{\mathrm{d}x} \\ f''(x) &= 6x + 2a \end{aligned}$$

and

$$
\begin{aligned}
y^2 &= x^3 + ax^2 + bx + c \\
\implies 2y\frac{\mathrm{d}y}{\mathrm{d}x} &= 3x^2 + 2ax + b \\
\implies \frac{\mathrm{d}y}{\mathrm{d}x} &= \frac{3x^2 + 2ax + b}{2y} \\
\implies \frac{\mathrm{d}^2y}{\mathrm{d}x^2} &= \frac{(6x + 2a)2y - 2\frac{\mathrm{d}y}{\mathrm{d}x}(3x^2 + 2ax + b)}{4y^2} \\
&= \frac{f''(x)2y - \frac{f'(x)}{y}f'(x)}{4f(x)} \\
&= \frac{f''(x)2y^2 - f'(x)^2}{4yf(x)} \\
&= \frac{2f''(x)f(x) - f'(x)^2}{4yf(x)} \\
&= \frac{\psi_3(x)}{4yf(x)}.
\end{aligned}
$$

Certainly, P is of order 1 if and only if $P = \mathcal{O}$ since $\mathcal{O}$ is the identity, so $P \neq \mathcal{O}$ if P is of order 3. By definition, points of inflection are points where the second derivative is equal to 0, and the expression above is equal to 0 if and only if $\psi_3(x) = 0$ and $y \neq 0$ (since $y = 0$ if and only if $f(x) = 0$). Indeed, if y were 0, then P would be of order 2, not 3, so we can omit the restriction on y and determine that the points of order 3 are exactly the points of inflection which are not $\mathcal{O}$.

(b) Note that if $a = b = c = 0$, then there is exactly one distinct real root, so we will prove the hypothesis at least for *most* cases. Importantly, when $a = b = c = 0$, the curve is singular, so perhaps the other cases that this may not work for are also cases where the curve is singular. Given that we have not discussed any criteria for determining whether a curve is singular, we will simply ignore edge cases such as this one after acknowledging them.

First, recall that

$$
\begin{aligned}
\psi_3(x) &= 3x^4 + 4ax^3 + 6bx^2 + 12cx + 4ac - b^2 \\
&= 2f(x)f''(x) - f'(x)^2
\end{aligned}
$$

and note that

$$
\begin{aligned}
f(x) &= x^3 + ax^2 + bx + c \\
f'(x) &= 3x^2 + 2ax + b \\
f''(x) &= 6x + 2a,
\end{aligned}
$$

so $f''(x) = 0$ if and only if $x = -a/3$. At this same x-value, we have

$$\begin{aligned} f'(-a/3) &= 3(-a/3)^2 + 2a(-a/3) + b \\ &= \frac{a^2}{3} - \frac{2a^2}{3} + b \\ &= \frac{3b - a^2}{3}. \end{aligned}$$

Thus $f'(-a/3) = 0$ whenever $3b = a^2$. We will ignore this case because to handle it using this method, we would need to find general roots of $f(x)$, which we determine is beyond the scope of this course.

Thus, if $3b \neq a^2$,

$$\psi_3(-a/3) = 2f(-a/3)f''(-a/3) - f'(-a/3)^2 = 0 - f'(-a/3)^2 < 0.$$

Since ψ_3 is a quartic with positive leading coefficient,

$$\lim_{x \to -\infty} \psi_3(x) = \lim_{x \to \infty} \psi_3(x) = \infty,$$

so the Intermediate Value Theorem tells us that ψ_3 has at least two real roots.

It is not clear to me how to show that there are exactly two roots. My only idea is to turn to finding roots of $\psi_3'(x)$:

$$\psi_3'(x) = 12x^3 + 12ax^2 + 12bx + 12c = 12f(x).$$

Thus $\psi_3'(x) = 0$ whenever $f(x) = 0$, which we know occurs at either 1 or 3 values. By graphing, it seems that ψ_3' can in fact have 1, 2, or 3 distinct roots. If it were always one root, then we would have shown the graph could never "go below the x-axis" a second time, restricting the number of roots of ψ_3 to two, but this is not the case.

Continuing on, then, we show that

□

3. $*$ Construct yourself an elliptic curve with as high rank as you can. For MTH 346 students, full credit will be awarded for a curve of rank ≥ 4. For MTH 646 students, full credit will be awarded for a curve of rank ≥ 5.

You should use Magma to test the rank of your elliptic curve.

When I say construct such a curve yourself, I mean you should *not* just look up curves in databases of curves with high rank. You should use some method to search. Just testing all curves in a box (say those of the form $y^2 = x^3 + Ax + B$ where $A, B \in \mathbb{Z}$ and $0 \leq A, B \leq 100$) will likely work for finding a rank 4 curve. For rank ≥ 5 you need a way to decrease the number of curves you are testing, or find a systematic way to construct curves with lots of rational points on them.

Proof. Here's a rank 4 for good measure: $y^2 = x^3 + 2x + 438$.

And then here's a rank 5: $y^2 = x^3 + 364x + 1961$. And a second one with relatively similar coefficients: $y^2 = x^3 + 371x + 1156$.

I fully admit, I found these by searching in a box, but I did find them! I searched EllipticCurve([a,b]) with $a, b \in [1, 2000]$. □